

FACTSHEET FS09 · 52-week high

Do stocks near their high keep going?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY

Anchoring momentum

ORIGIN

George & Hwang (2004)

FAMILY

Momentum & trend

UNIVERSES

US · EU · UK · DK · SCANDI ·
World (point-in-time)

BACKTEST

Feb 2013 – Aug 2026, monthly, net
of costs

BOOKS

Long/short and long-only (stocks
closest to their 52-week high)

REBALANCE

Monthly

STATUS

Descriptive factsheet, not a
registered trial

The claim. Stocks trading close to their 52-week high outperform those far below it, and this nearness explains much of momentum. George & Hwang (2004) found it for US stocks 1963–2001; investors anchor on the high and under-react to good news near it. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -5.9% to +5.8% a year and was positive in 2 of the six; no universe passes the 2.87 gate. The long-only book (stocks closest to their 52-week high) had a higher Sharpe than the equal-weight universe in 5 of the six (EU, UK, DK, SCANDI, World), with alphas of -0.1% to +6.1%, passing the gate in EU, DK. **What goes wrong (§10):** the main drags are a market bet (average beta -0.61, worth -8.0% a year in 2013–19), costs of about 2.1% a year. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Momentum & trend; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours 55-day breakout, MIN5; after its closest neighbours and the market it keeps an alpha of -2.1% to +3.6% (largest |t| 2.6), so it carries some information of its own.

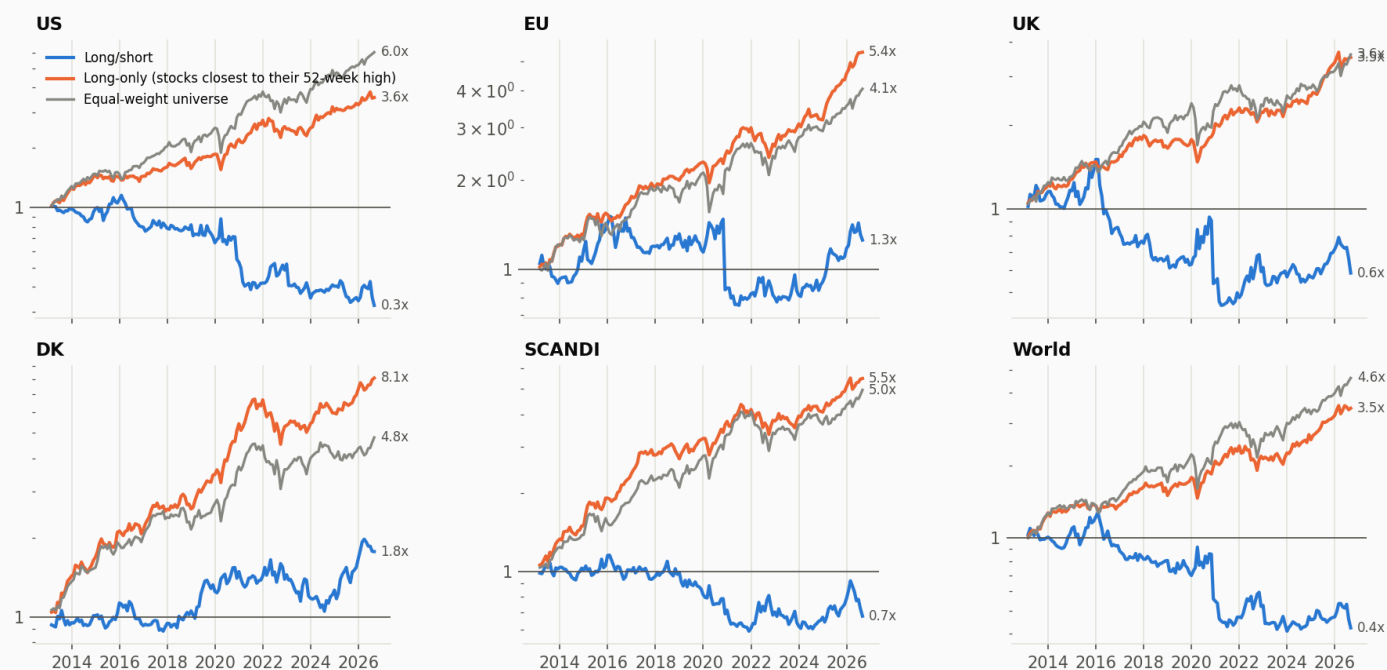
1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	501 / 10	-5.9%	-1.19	+5.3% (1.4)	-0.77	+9.8%	0.79	-20%	+14.1%	0.95	+15.3%
EU	238 / 10	+4.3%	0.84	+15.2% (4.0)	-0.95	+13.2%	1.21	-21%	+10.9%	0.78	+10.1%
UK	239 / 10	-0.8%	-0.13	+9.6% (2.0)	-1.00	+9.7%	0.93	-20%	+9.9%	0.74	+8.5%
DK	19 / 3	+5.8%	1.39	+11.1% (2.8)	-0.42	+16.6%	1.09	-33%	+12.3%	0.82	+10.3%
SCANDI	62 / 5	-1.6%	-0.44	+5.4% (1.7)	-0.55	+13.3%	1.05	-23%	+12.5%	0.91	+10.7%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
World	971 / 10	-4.0%	-0.78	+6.5% (1.9)	-0.84	+9.6%	0.80	-21%	+11.9%	0.80	+10.8%

Monthly, local currency (World in USD). L/S = equal-weighted stocks closest to their 52-week high minus stocks furthest below it, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

52-week high on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjernerig - Project 10 factsheets

2. Strategy description

Stocks trading close to their 52-week high outperform those far below it, and this nearness explains much of momentum. George & Hwang (2004) found it for US stocks 1963–2001; investors anchor on the high and under-react to good news near it.

Why it might work. Anchoring: traders use the 52-week high as a reference and are reluctant to bid prices above it, so good news is absorbed slowly (George & Hwang 2004).

Rule	Original (George & Hwang (2004))	This factsheet
Signal	Price / 52-week high	Total-return price / 252-day high at month-end t
Portfolio	Top 30% minus bottom 30%, 6-month holding	Extreme quantile L/S, one-month holding
Weighting	Equal weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** We hold for one month; George & Hwang hold for six. A longer holding period would cut turnover.

4. Signal creation

$HI52_{i,t} = P_{i,t} / \max(P_{i,t-251..t})$. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = stocks closest to their 52-week high minus stocks furthest below it; **long-only** = stocks closest to their 52-week high; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.

- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 501 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-5.9%	-8.0%	21.5%	-0.28 [-0.71, 0.15]	-0.47	-72%	47%	-1.19	-0.77	+5.3% (1.4)	16.5%
Long-only (net)	+10.2%	+9.8%	12.9%	0.79 [0.37, 1.25]	1.26	-20%	58%	3.97	0.72	-0.1% (-0.1)	7.6%
Stocks closest to their 52-week high (gross)	+12.0%	+11.7%	12.9%	0.93 [0.50, 1.40]	1.55	-19%	60%	4.63	0.71	+1.7% (1.1)	7.4%
Stocks furthest below it (gross)	+14.4%	+11.7%	26.0%	0.56 [0.13, 1.09]	0.73	-43%	61%	2.50	1.49	-7.1% (-2.4)	14.6%
EW universe	+14.4%	+14.1%	15.1%	0.95 [0.52, 1.53]	1.52	-25%	68%	4.79	1.00	—	8.9%
Size-weighted universe	+15.4%	+15.3%	14.4%	1.07 [0.61, 1.64]	1.79	-24%	69%	5.27	0.90	+2.4% (1.8)	8.3%

EU (11 national blue-chip indices, point-in-time), median 238 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+4.3%	+1.7%	21.8%	0.20 [-0.23, 0.73]	0.10	-50%	54%	0.84	-0.95	+15.2% (4.0)	14.3%
Long-only (net)	+13.0%	+13.2%	10.8%	1.21 [0.66, 1.85]	2.20	-21%	64%	4.63	0.61	+6.1% (3.9)	5.8%
Stocks closest to their 52-week high (gross)	+14.6%	+15.0%	10.8%	1.36 [0.80, 2.01]	2.58	-20%	66%	5.17	0.61	+7.7% (4.8)	5.7%
Stocks furthest below it (gross)	+7.3%	+4.1%	26.0%	0.28 [-0.17, 0.70]	0.27	-41%	50%	1.16	1.56	-10.5% (-3.7)	13.0%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Stocks closest to their 52-week high (gross)	+11.6%	+11.4%	12.4%	0.94 [0.48, 1.48]	1.52	-20%	63%	4.46	0.68	+3.1% (2.4)	7.6%
Stocks furthest below it (gross)	+12.5%	+9.5%	26.3%	0.47 [0.03, 0.97]	0.60	-43%	56%	1.96	1.52	-6.6% (-2.7)	14.1%
EW universe	+12.5%	+11.9%	15.7%	0.80 [0.33, 1.38]	1.20	-29%	64%	3.53	1.00	—	9.4%
Size-weighted universe	+11.4%	+10.8%	15.0%	0.76 [0.29, 1.33]	1.14	-28%	63%	3.23	0.94	-0.4% (-0.5)	9.0%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	-2.0%	-7.2%	+15.3%	-4.9%	+3.9%	+0.9%
2014	+0.8%	+25.3%	+0.6%	+6.4%	-2.4%	+8.7%
2015	+11.3%	+22.5%	+29.7%	+11.7%	+13.7%	+11.6%
2016	-27.4%	-10.3%	-45.9%	-18.1%	-12.8%	-27.9%
2017	-1.4%	-6.7%	-7.9%	+2.1%	-4.5%	-7.5%
2018	+4.2%	+10.3%	-10.2%	+11.5%	+3.5%	-1.2%
2019	-18.8%	-13.2%	-6.4%	+18.6%	-17.9%	-13.4%
2020	-20.2%	-24.3%	-13.5%	+19.0%	-12.7%	-21.8%
2021	-21.4%	-4.3%	-12.3%	-4.0%	-14.8%	-14.0%
2022	+24.6%	-3.1%	+28.4%	+1.3%	+15.4%	+11.7%
2023	-32.5%	+2.0%	-7.5%	-18.6%	-11.7%	-21.8%
2024	+14.6%	+10.9%	-0.0%	+7.4%	+8.2%	+11.6%
2025	-15.5%	+29.8%	+26.1%	+31.9%	+18.1%	+2.8%
2026	-5.1%	+6.6%	-18.1%	+6.2%	-14.5%	-10.0%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+26.0%	+21.3%	+22.6%	+43.2%	+33.3%	+27.2%
2014	+14.9%	+7.6%	+4.1%	+17.9%	+12.3%	+7.3%
2015	-2.2%	+17.2%	+15.6%	+25.7%	+27.7%	-0.2%
2016	+1.4%	+15.2%	+5.9%	+1.1%	+24.4%	+0.6%

Year	US	EU	UK	DK	SCANDI	World
2017	+11.2%	+7.7%	+17.5%	+19.7%	+17.4%	+19.0%
2018	-1.7%	+5.1%	-10.1%	+0.2%	-3.3%	-7.9%
2019	+18.1%	+15.3%	+7.6%	+35.3%	+19.7%	+18.8%
2020	+18.6%	+9.1%	+14.1%	+55.7%	+16.2%	+17.0%
2021	+24.8%	+19.6%	+13.9%	+23.3%	+11.1%	+15.6%
2022	-9.9%	-13.3%	-3.7%	-20.0%	-9.7%	-11.6%
2023	+1.5%	+18.9%	+6.8%	+1.6%	+4.6%	+5.8%
2024	+21.1%	+3.1%	+7.3%	+9.6%	+3.4%	+13.4%
2025	+7.8%	+43.5%	+32.2%	+23.4%	+25.3%	+26.2%
2026	+8.8%	+17.9%	+4.2%	+10.1%	+7.3%	+7.1%

7. Trading record

The rebalance log per universe is in `results/xs_FS09_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	51 / 50	73%	28%	47%	-23.2%	2020-04
EU	163	24 / 23	66%	22%	54%	-42.2%	2020-11
UK	163	24 / 23	69%	23%	58%	-40.7%	2020-11
DK	163	7 / 6	30%	21%	55%	-12.9%	2023-11
SCANDI	163	13 / 12	48%	24%	47%	-12.6%	2020-11
World	163	98 / 97	71%	26%	51%	-34.8%	2020-11

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks closest to their 52-week high	Short: stocks furthest below it
US	MPC, SLB, VLO, DINO, CRWD, PSX, GEN, UNM	CSGP, FISV, PODD, APP, BSX, ORCL, COIN, ZTS
EU	SKF-B.ST, RAND.AS, PUIG.MC, PZU.WA, ABN.AS, DSFIR.AS, INGA.BR, JYSK.CO	ONTEX.BR, FCT.MI, AVIO.MI, STLAM.MI, AGFB.BR, STLAP.PA, NYR.BR, UMG.AS
UK	VCT.L, AAL.L, ASL.L, RS1.L, CCC.L, BRWM.L, BUT.L, BYIT.L	AML.L, VTY.L, CWR.L, TEP.L, RPI.L, DATA.L, OXB.L, SMWH.L

Universe	Long: stocks closest to their 52-week high	Short: stocks furthest below it
DK	JYSK.CO, NDA-DK.CO, NSIS-B.CO, PNDORA.CO, VWS.CO, DANSKE.CO, ISS.CO, DEMANT.CO	ZEAL.CO, AMBU-B.CO, ORSTED.CO, RBREW.CO, DSV.CO, COLO-B.CO, NOVO-B.CO, GN.CO
SCANDI	KESKOB.HE, JYSK.CO, SKF-B.ST, NDA-SE.ST, NDA-DK.CO, NSIS-B.CO, NDA-FI.HE, SHB-A.ST	NOKIA.HE, ORSTED.CO, RBREW.CO, QTCOM.HE, WRT1V.HE, AMBU-B.CO, DSV.CO, ERIC-B.ST

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-9.9%	-5.58	0.84	size +0.25 (+2.2), value -0.41 (-2.4), momentum +1.04 (+5.7), low_risk +1.28 (+6.3), quality +0.48 (+2.1), short_term_reversal -1.14 (-6.5)
EU	French Europe 5F + WML	-5.3%	-1.54	0.64	Mkt-RF -0.42 (-4.6), RMW +0.58 (+2.4), WML +1.21 (+6.8)
UK	JKP GBR 7 themes	-6.8%	-1.55	0.76	momentum +1.08 (+7.6), low_risk +1.57 (+5.2), quality +0.80 (+2.9), short_term_reversal -0.78 (-4.0)
DK	JKP DNK 7 themes	-3.5%	-1.30	0.58	momentum +1.04 (+10.6), low_risk +0.59 (+4.6)
SCANDI	French Europe 5F + WML	-9.3%	-3.02	0.44	Mkt-RF -0.23 (-3.2), WML +0.85 (+9.2)
World	JKP World 7 themes	-5.0%	-1.25	0.70	mkt -0.15 (-2.0), value -0.34 (-2.5), momentum +0.96 (+8.2), low_risk +0.92 (+5.5), short_term_reversal -0.91 (-4.0)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-1.3%	-0.97	0.86	mkt +0.94 (+30.0), momentum +0.34 (+3.5), low_risk +0.39 (+4.5), quality +0.25 (+2.4), short_term_reversal -0.36 (-3.7)
EU	French Europe 5F + WML	+5.7%	2.61	0.56	Mkt-RF +0.53 (+11.4), HML +0.29 (+2.5), WML +0.23 (+4.0)
UK	JKP GBR 7 themes	+3.9%	1.90	0.56	mkt +0.44 (+11.1), value +0.17 (+2.0), momentum +0.33 (+4.1), quality +0.41 (+3.7)
DK	JKP DNK 7 themes	+5.3%	1.85	0.66	mkt +0.75 (+16.9), value +0.17 (+2.8), momentum +0.31 (+3.7), quality +0.23 (+2.4)
SCANDI	French Europe 5F + WML	+5.9%	2.22	0.50	Mkt-RF +0.60 (+11.0), RMW +0.39 (+2.7), WML +0.21 (+2.6)
World	JKP World 7 themes	+4.1%	1.70	0.67	mkt +0.67 (+13.5), momentum +0.30 (+3.2), low_risk +0.31 (+2.8), quality +0.23 (+2.1)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books $\times$ 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in 2 of the six; passes in 0 of 6.
- **L/S CAPM alpha:** +5.3% to +15.2%, $|t|$ past the gate in 1 of 6.
- **Long-only:** Sharpe above the equal-weight universe in 5 of the six; alpha -0.1% to +6.1%, passing in 2 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.15, EU 0.75, UK 0.45, DK 0.89, SCANDI 0.35, World 0.23.
- **Publication decay:** gross L/S -1.8% to +6.7% a year in 2013–19 against -3.9% to +9.1% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	-1.1%	-0.65	-9.2%	+8.1%	-2.5%	-2.8%	+1.7%	104%	2015-04 -13%, 2019-01 -12%, 2019-06 -11%
EU	+6.7%	-0.94	-10.9%	+17.7%	-2.1%	+2.5%	+4.3%	87%	2015-02 -13%, 2016-04 -11%, 2016-03 -11%
UK	-1.8%	-0.74	-9.7%	+8.0%	-2.3%	-2.8%	+1.0%	95%	2016-02 -19%, 2016-04 -14%, 2015-02 -11%
DK	+6.4%	-0.16	-2.5%	+8.8%	-1.3%	+4.6%	+1.8%	56%	2017-08 -8%, 2013-08 -7%, 2016-08 -7%
SCANDI	+0.6%	-0.44	-6.8%	+7.4%	-1.8%	+3.4%	-2.8%	77%	2018-07 -9%, 2013-08 -8%, 2019-01 -8%
World	-0.7%	-0.75	-9.1%	+8.4%	-2.4%	-1.7%	+1.0%	99%	2016-04 -10%, 2019-01 -10%, 2015-04 -10%

On average across the six universes the gross spread was +1.7% a year, of which the market exposure (beta -0.61) contributed -8.0%; the beta-adjusted spread (CAPM alpha) was +9.7%. Costs took 2.1% a year at 86% monthly turnover across both legs. The long leg beat the universe by +0.5% and the short leg lagged it by +1.2% a year, so most of the spread

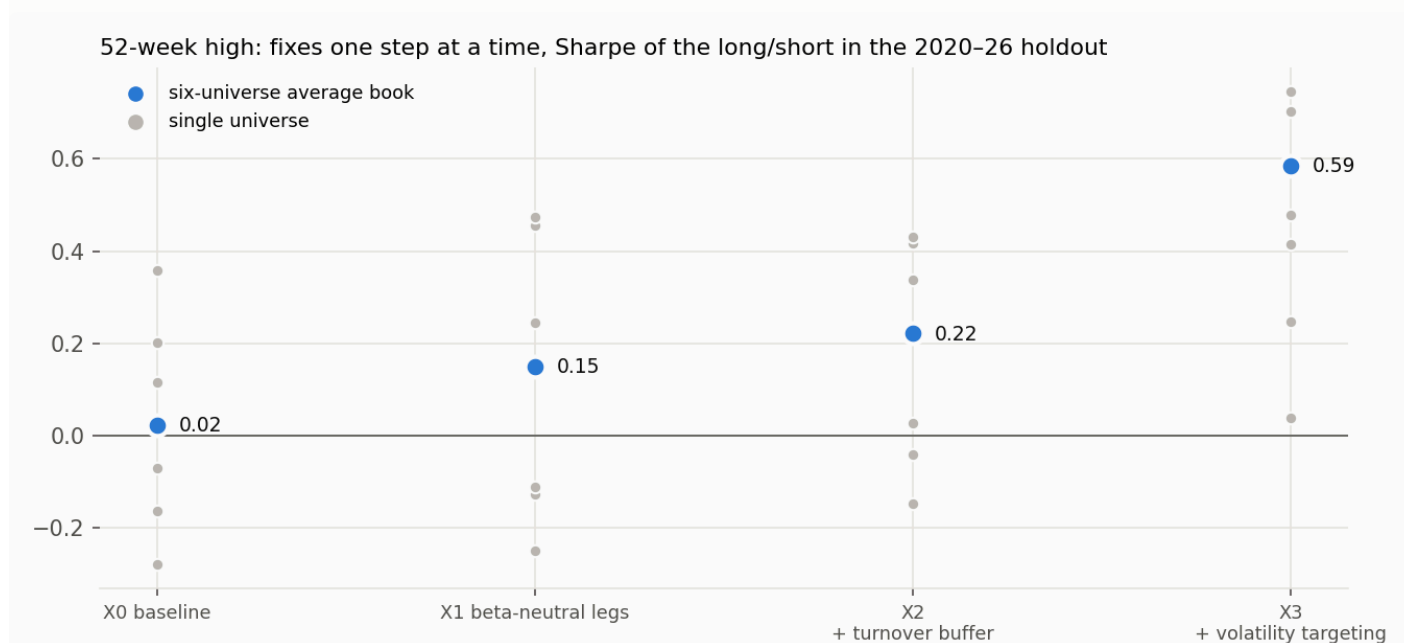
comes from the short side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +1.9%, +3.4%, -0.3% a year.

10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	-0.07	—	0.02	—	—	+0.5%	-39%	+10.0% (2.3)	2.0%
X1 beta-neutral legs	0.22	+0.40 (p 0.001)	0.15	+0.13 (p 0.203)	5 of 6	+2.2%	-33%	+5.4% (1.2)	2.3%
X2 + turnover buffer	0.37	+0.20 (p 0.006)	0.22	+0.07 (p 0.144)	4 of 6	+2.8%	-27%	+5.6% (1.4)	1.2%
X3 + volatility targeting	0.28	-0.09 (p 0.582)	0.59	+0.36 (p 0.001) ✓	6 of 6	+6.0%	-17%	+8.0% (2.3)	1.2%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.

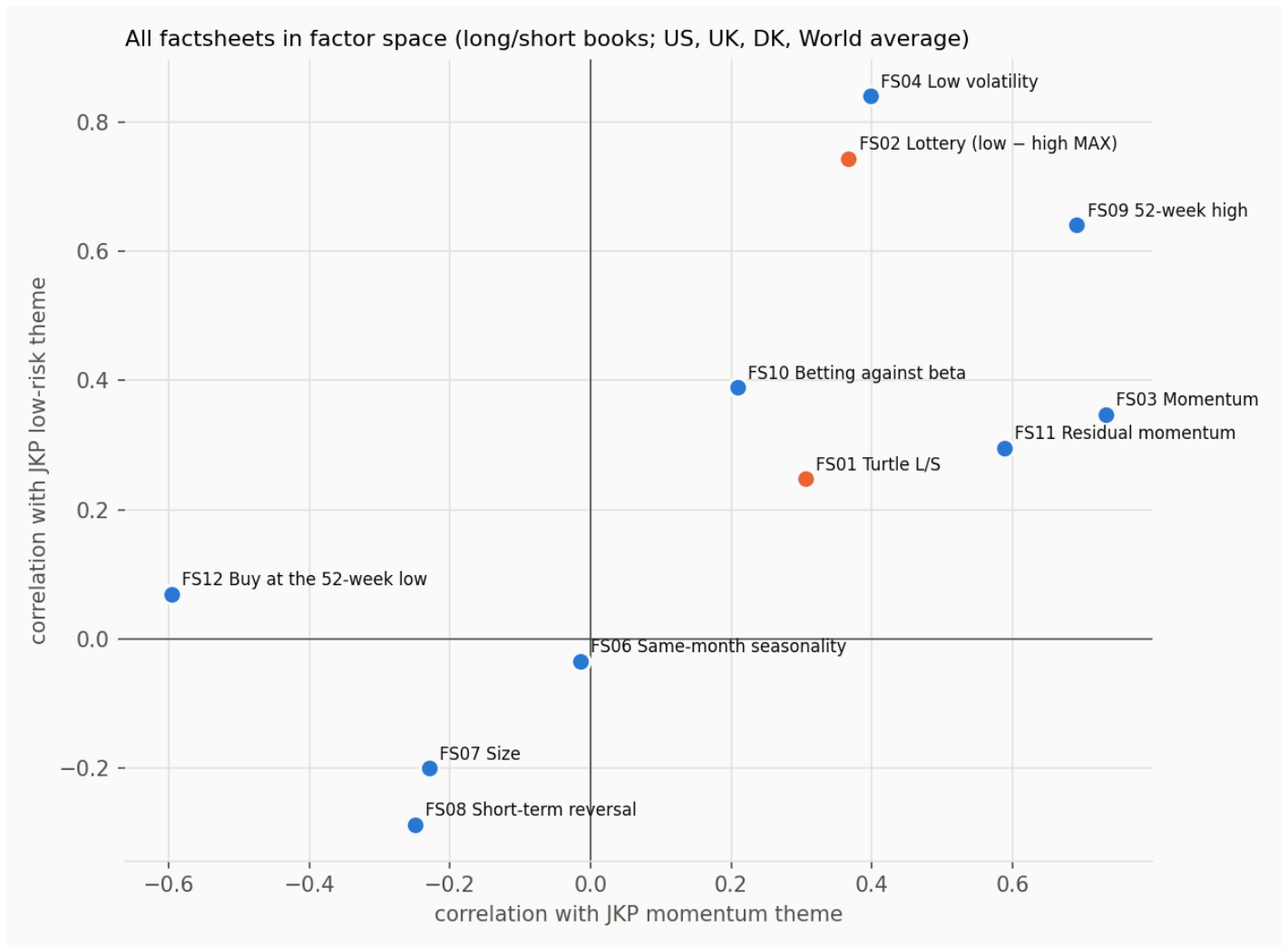


- **X1 beta-neutral legs:** +0.40 design, +0.13 holdout; helps in both windows but does not pass the gate.
- **X2 + turnover buffer:** +0.20 design, +0.07 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.09 design, +0.36 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe 0.02 → 0.59, return +0.5% → +6.0% a year (six-universe book).

11. Classification: where does it fit?

Dimension	52-week high
Series family	Momentum & trend
Academic style	Anchoring momentum
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 86% of the two legs replaced monthly
Market exposure	L/S beta -1.00 to -0.42
Payoff shape	Treynor–Mazuy γ -4.37 to 1.22 (t -3.0 to 1.2); worst 10% of market months +4.0% to +5.9% a month, best 10% -8.5% to -2.6%
Nearest factor theme	momentum (2 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	55-day breakout (6 universes), MIN5 (4 universes), Volatility (3 universes)
Economic rationale	Anchoring: traders use the 52-week high as a reference and are reluctant to bid prices above it, so good news is absorbed slowly (George & Hwang 2004).
Publication	George & Hwang 2004
Where it fits	In the momentum cluster of the library; with some information beyond its neighbours.

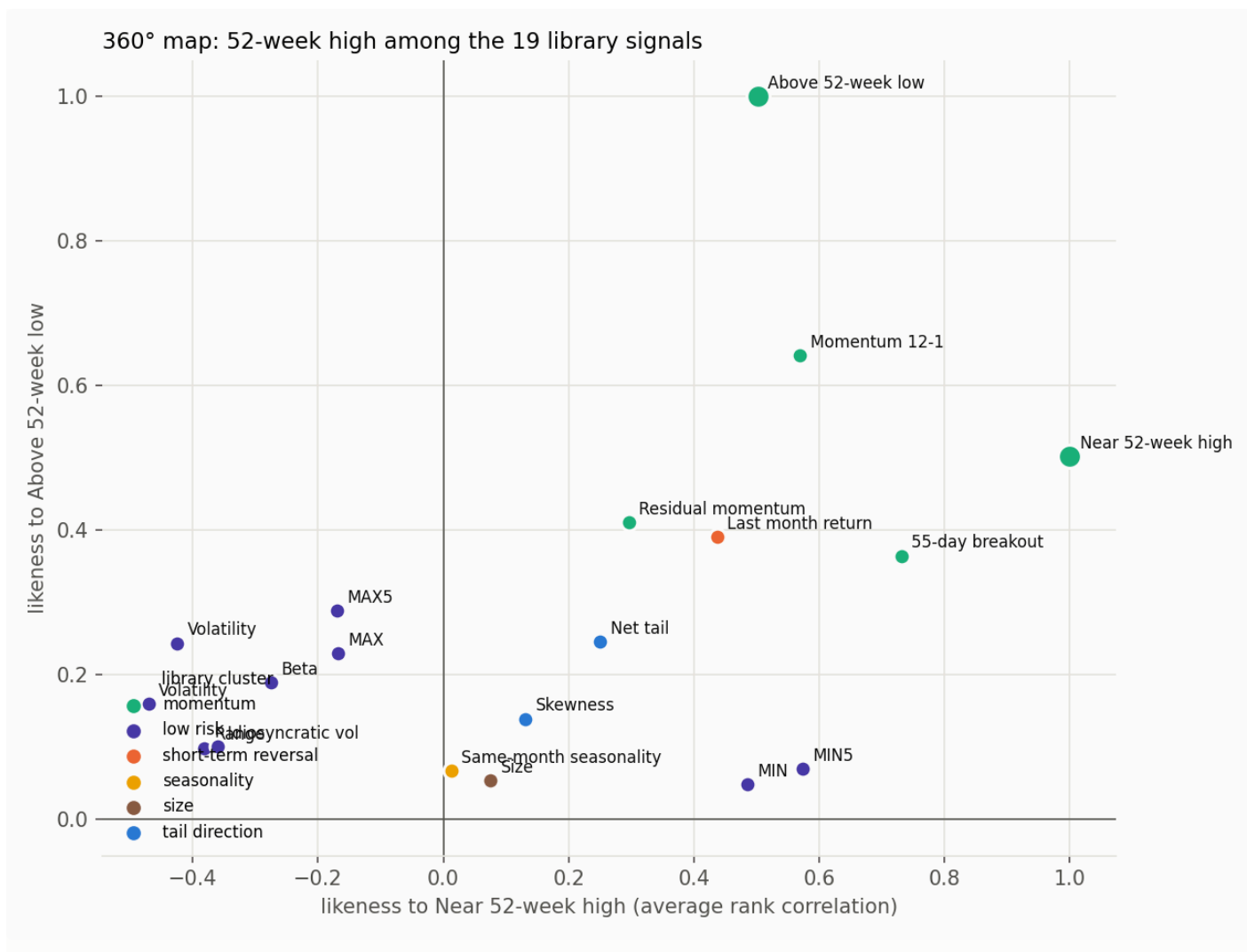


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Above 52-week low**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Near 52-week high (this strategy)	momentum	+1.00	+1.00	+2.5%	+8.3% to +18.8%	6 / 0
55-day breakout	momentum	+0.73	+0.83	+0.7%	+5.6% to +12.6%	3 / 0
Momentum 12-1	momentum	+0.57	+0.77	+7.7%	+3.9% to +17.8%	5 / 0
MIN5	low risk	+0.57	+0.76	-2.0%	+2.4% to +13.2%	4 / 0
Volatility	low risk	-0.47	-0.74	+3.0%	-10.3% to -3.7%	0 / 5
Volatility (252 days)	low risk	-0.43	-0.70	+5.1%	-10.3% to -0.2%	0 / 1

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
MIN (worst day)	low risk	+0.48	+0.70	-1.5%	+2.9% to +11.0%	3 / 0
Range (MAX–MIN)	low risk	-0.38	-0.69	+1.2%	-11.3% to -2.3%	0 / 4
Beta	low risk	-0.28	-0.65	+5.5%	-15.7% to -2.6%	0 / 3
Idiosyncratic vol	low risk	-0.36	-0.58	-1.5%	-10.4% to -0.8%	0 / 3
MAX (best day)	low risk	-0.17	-0.55	+1.9%	-9.3% to +1.0%	0 / 2
MAX5	low risk	-0.17	-0.52	+2.6%	-8.5% to +0.4%	0 / 2
Last month return	short-term reversal	+0.44	+0.51	+0.9%	-3.2% to +8.4%	0 / 0
Residual momentum	momentum	+0.30	+0.47	+6.1%	+7.3% to +13.0%	5 / 0
Above 52-week low (mirror)	momentum	+0.50	+0.46	+8.8%	+4.4% to +13.8%	4 / 0
Size	size	+0.07	+0.31	-1.5%	-3.8% to +4.6%	0 / 0
Net tail (MAX+MIN)	tail direction	+0.25	+0.17	-1.0%	-5.3% to +4.2%	0 / 0
Same-month seasonality	seasonality	+0.01	+0.08	-1.8%	-8.5% to +7.2%	0 / 0
Skewness	tail direction	+0.13	-0.03	-0.2%	-5.4% to +4.5%	0 / 1



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with $ t \geq 2$
US	-1.6%	-1.22	0.94	55-day breakout +0.54 (+10.6), Momentum 12-1 +0.43 (+18.7), Volatility -0.24 (-6.2)
EU	+3.6%	2.61	0.95	MIN5 +0.26 (+4.3), 55-day breakout +0.33 (+9.0), Momentum 12-1 +0.43 (+14.2), Volatility -0.22 (-3.2)
UK	+1.9%	0.62	0.87	55-day breakout +0.53 (+7.6)
DK	+0.5%	0.18	0.73	Momentum 12-1 +0.35 (+4.9), 55-day breakout +0.25 (+2.6), Above 52-week low +0.27 (+3.3), Volatility -0.35 (-4.1)
SCANDI	+2.0%	1.10	0.81	55-day breakout +0.44 (+5.9), Momentum 12-1 +0.42 (+10.1), Volatility -0.23 (-2.7)
World	-2.1%	-2.37	0.96	55-day breakout +0.48 (+7.5), Momentum 12-1 +0.46 (+13.9), Volatility -0.15 (-3.3)

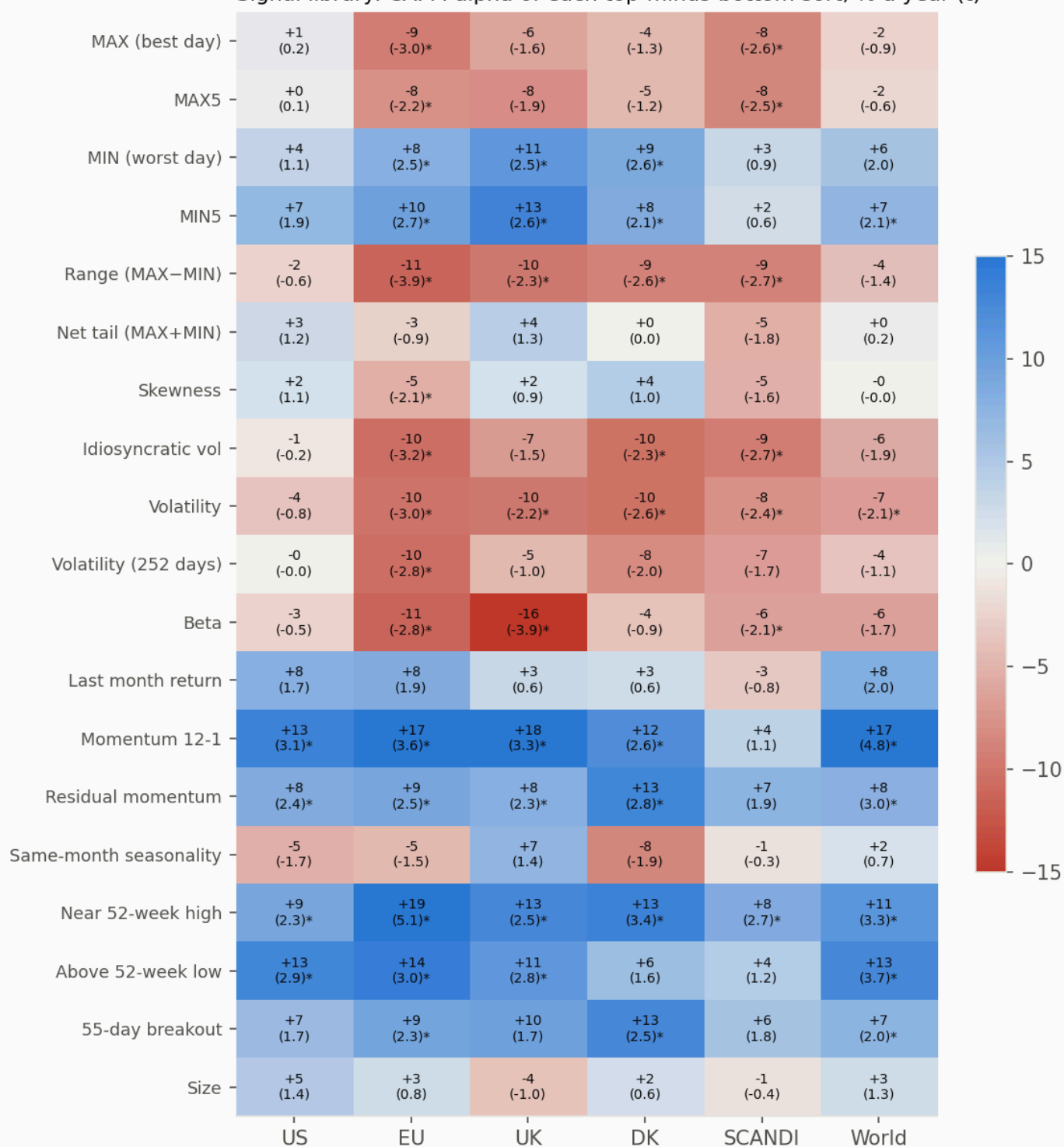
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Near 52-week high -	10.6%	12.8%	14.1%
middle -	8.6%	12.1%	15.5%
high Near 52-week high -	11.2%	12.3%	14.7%
	low Above 52-week low	middle	high Above 52-week low

Holding Above 52-week low fixed (down a column), moving from low to high Near 52-week high changes the return by +0.6, -0.5 and +0.6 points a year; holding Near 52-week high fixed (along a row), moving from low to high Above 52-week low changes it by +3.5, +6.9 and +3.6. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -15% vs +34% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +2.5%, +1.9%, +3.4%, -0.3% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Data gap.** We hold for one month; George & Hwang hold for six. A longer holding period would cut turnover.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

14. Academic references

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15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS09 → `build_xs.py`
FS09. Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.